

ESPHome AUTOMATION

SOMNATH BERA

ESPHome is an open source system for macOS, Linux, and Windows, comprising two primary components:

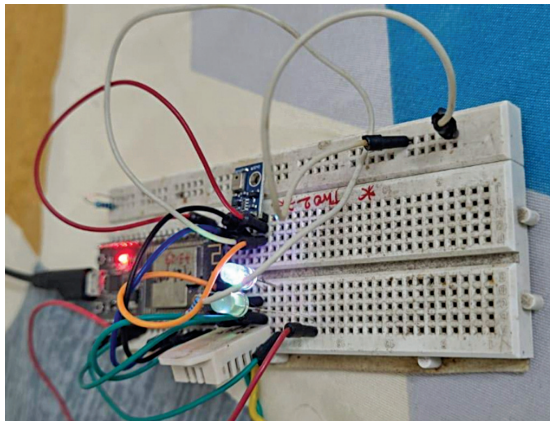


Fig. 1: Author's prototype

Bluetooth proxy for Bluetooth low energy (BLE) sensor integration and ESPHome Automation for Wi-Fi-based smart home devices.

It supports micro-controllers like ESP32, ESP32-S2/S3/C3, ESP8266, and the Raspberry Pi Pico W, though the Bluetooth Proxy feature is exclusive to ESP32. The Pi Pico W currently faces limitations, as it is not displayed on the ESPHome dashboard or the browser.

This system turns microcontrollers into efficient home automation

tools, enabling simple sensor and actuator connections with minimal programming. Through an intranet browser, users can control LED brightness, toggle switches, manage buzzers, monitor gas sensors like

BILL OF MATERIALS

Components	Quantity
ESP32 development board (MOD1)	1
DHT22 sensor (S2)	1
ATH10 sensor (S1)	1
Power supply: 5V regulated DC	1
Resistor: 220Ω (R1, R2)	2
Resistor: 4.7kΩ (R3)	1
LED1, LED2	2
MQ135 sensor (optional)	1

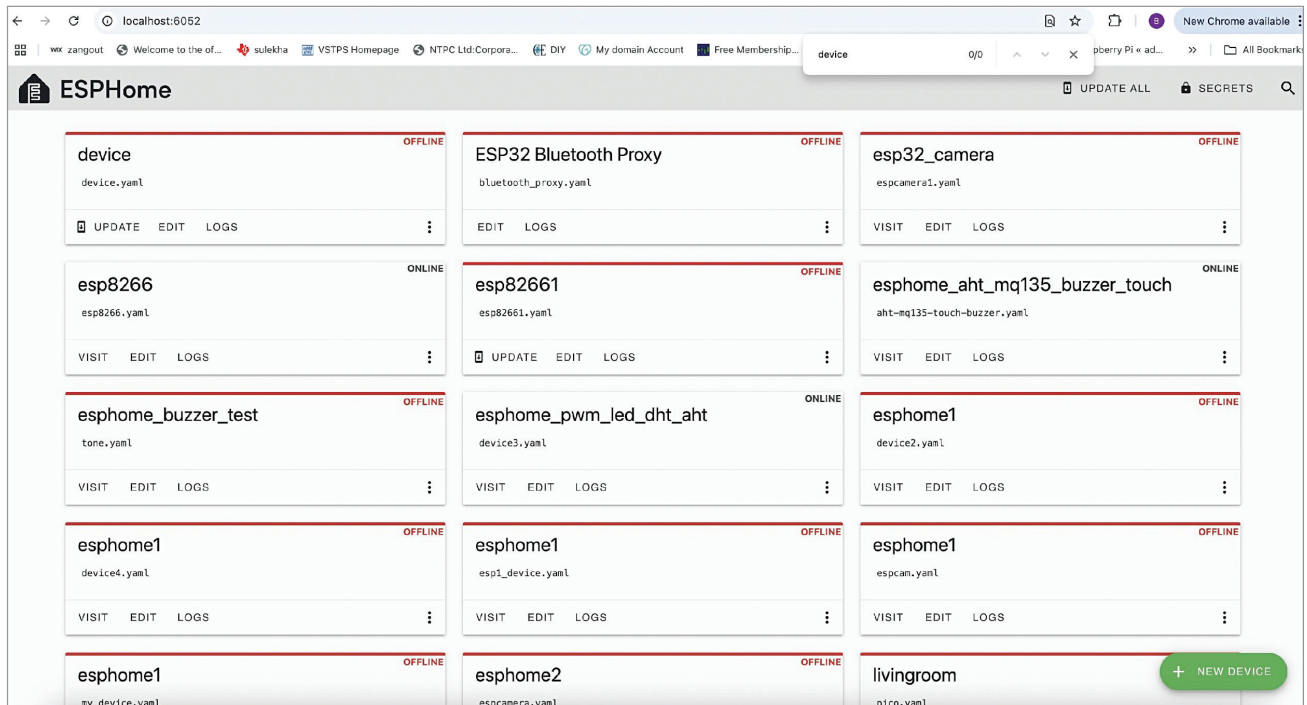


Fig. 2: ESPHome dashboard